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1. Introduction

Etymology currently lacks algorithmic structure. While the comparative method provides robust frameworks for sound change (Grimm 1822, Verner 1875, Grassmann 1863), connecting modern words to ancestral roots remains interpretive. Standard practice proceeds forward: given a PIE root, scholars trace reflexes through daughter languages. The inverse problem—given a modern reflex, identify the source root—has no formalized procedure. We propose that consonantal skeletons provide sufficient structural information to reverse-engineer etymologies systematically. Stripping affixation and vocalic material from any Indo-European word yields a CVC (consonant-vowel-consonant) frame that maps to a finite set of PIE roots. Progressive filtering through vowel grade, morphological context, and cross-linguistic attestation narrows this set mechanically.

This paper formalizes that process as a six-step protocol, validates it against inscriptional evidence, and demonstrates that the method operates entirely within established IE linguistic principles. No novel theoretical claims are introduced. We simply systematize what historical linguists already do implicitly.

2. Theoretical Foundations

The Barnaby Protocol employs three established frameworks:

First, sound correspondences as formalized by Grimm (1822), with systematic refinements by Verner (1875), Grassmann (1863), and Bartholomae (1883). These laws specify how PIE phonemes evolve in each daughter branch with predictable regularity.

Second, ablaut patterns as described by Brugmann (1876). PIE roots show systematic vowel alternations: E-grade (full vowel), O-grade (result/perfect), Zero-grade (reduced), and lengthened forms (intensive). These alternations carry grammatical function and appear consistently across the IE family.

Third, morphological reconstruction as established by Brugmann and Delbrück (1886-1900).

PIE morphology follows regular templates: root plus thematic vowel plus derivational suffix plus inflectional ending. Each layer is identifiable and removable through systematic analysis.

The protocol applies these frameworks in reverse sequence. Where traditional etymology moves from PIE root to modern reflex, we move from modern reflex to PIE root using the same laws in reverse.

3. The Protocol

The Barnaby Protocol consists of six sequential steps. Each applies established IE principles to narrow the candidate space.

Step 1: Affix Stripping

Remove all non-root morphology using branch-specific paradigms. This includes inflectional endings (case, number, person, tense), derivational suffixes (word-class changers), and thematic vowels (stem formants). What remains is the consonantal skeleton: C₁VC₂.

Step 2: Skeleton Mapping

Query PIE root databases (Pokorny 1959, LIV² 2001, or equivalent) for all roots matching the skeleton pattern. This generates a finite candidate set, typically 20-80 roots depending on skeleton frequency. Apply branch-specific sound laws in reverse to account for phonological changes.

Step 3: Vowel Filtering

Use the root vowel to filter candidates by ablaut grade. Apply Brugmann's principles (1876) to determine which PIE vowel grade the attested vowel reflects. This typically reduces candidates by 60-70%.

Step 4: Morphological Filtering

Apply morphological and contextual information. Removed affixes indicate word class, grammatical function, and semantic domain. Inscriptional context further constrains possibilities. This step typically reduces candidates to 1-5 roots.

Step 5: Candidate Review

Examine remaining PIE roots in detail. Check standard reconstructions, cross-branch attestations, and semantic coherence with the inscriptional context.

Step 6: Reflex Verification

For each candidate, apply sound laws forward to generate predicted reflexes across IE branches. Compare predictions to attested forms. Candidates that generate the observed reflex pattern are confirmed.

4. Demonstration: ΛΕΓΩ (Greek)

We demonstrate the protocol on a single well-attested example to show each step in detail. Inscriptional context: The word ΛΕΓΩ appears in numerous Greek inscriptions from 500 BCE onward. It functions as a first-person singular present indicative verb meaning "I speak" or "I say."

Step 1: Affix Stripping (Smyth 1920)

Form: ΛΕΓΩ

The -Ω ending marks first person singular, present tense, active voice in the thematic conjugation. Following standard Greek verbal paradigms (Smyth 1920:383), we remove this inflection.

Remaining: ΛΕΓ-

No thematic vowel separates root from ending in this verb. The root consonants appear directly before the personal ending.

Skeleton extracted: L-G

Root vowel: E (from ΛΕΓ)

Step 2: Skeleton Mapping (Pokorny 1959:658)

We query PIE databases for all roots with the structure L-vowel-G. Pokorny (1959:658) lists 67 such roots, including:

- *leǵ- "gather, collect"
- *leǵ- "speak, say"

- *leg- “drip, trickle”
- *lewg- “bend, wind”
- *lug- “break”
- *leg- “lie down”

Plus 61 additional roots with various meanings.

To account for sound changes, we apply Greek sound laws in reverse. The Greek form ΛΕΓ preserves PIE *leǵ- directly. The palatal *ǵ (palatovelar) becomes Greek γ before front vowels (Sihler 1995:203).

Starting candidate pool: 67 roots

Step 3: Vowel Filtering (Brugmann 1876)

The root vowel in ΛΕΓ is E (epsilon). In Greek, epsilon typically reflects PIE *e (e-grade) in full-grade forms. Following Brugmann’s ablaut analysis (1876), present tense verbs normally show e-grade.

We filter the 67 candidates to those showing e-grade in present formations:

- *leǵ- with e-grade (matches)
- *leg- with e-grade (matches)
- *lewg- contracts to *leg- in some forms (possible)

Candidates showing different grades:

- *loug- (o-grade primary) - eliminated
- *lug- (zero-grade or u-root) - eliminated
- *log- (o-grade) - eliminated

Remaining after vowel filter: 19 roots

Step 4: Morphological Filtering

The morphological structure provides additional constraints. The -Ω ending identifies this as a present tense thematic verb. The inscriptional contexts show it functioning as a speech verb.

Removed ending: -Ω (1sg present active)

Word class: Primary verb (not denominative)

Semantic field required: Speech, communication, or closely related action

Transitivity: Transitive verb (takes direct object in attestations)

We filter the 19 remaining roots to those semantically appropriate:

- *leǵ- “gather, collect” → semantic extension to “gather words” = “speak”

(plausible)

- *leǵ- “speak, say” → direct match (if distinct root)
- *leg- “drip” → semantic mismatch for speech verb (eliminated)
- *lewg- “bend” → semantic mismatch (eliminated)

Morphological context further supports: This verb takes direct objects (accusative) in inscriptions, consistent with transitive “speak, say” or “gather” semantics.

Remaining after morphological filter: 3 roots (possibly variants of same *leǵ-)

Step 5: Candidate Review

The three remaining candidates are likely the same PIE root *leǵ- with semantic development visible across branches.

*PIE leǵ- “gather, collect; speak”

Cross-linguistic attestations (Pokorny 1959:658):

- Greek: λέγω (legō) “I speak, say”

- Latin: legō “I gather, read, choose”
- Gothic: -ligan “lay, collect”
- Old Irish: ·legaim “I read”

The semantic development “gather” → “gather words” → “speak” is well-attested cross-linguistically. The same root shows both concrete gathering (Latin) and speech (Greek) across branches.

The O-grade form *loǵ-os yields:

- Greek: λόγος (logos) “word, speech, reason”
- Latin: (borrowed as) locus “place” (different root), but cf. legatio “sending of words”

Single root identified: *leǵ-

Step 6: Reflex Verification (Sihler 1995)

We now apply sound laws forward from PIE *leǵ- to verify predicted reflexes match attestations.

*PIE leǵ- → Proto-Greek

Applying sound changes per Sihler (1995):

- PIE *leǵ- maintains *l initially (no change)
- PIE *e remains e in Greek (no raising)
- PIE *ǵ (palatovelar) → Greek γ before e, i (Sihler 1995:203)

Predicted Proto-Greek form: *leg-

*Proto-Greek leg- → Attic Greek

No further changes affect this root in Attic:

- Initial *l → λ (unchanged)
- Stem vowel *e → ε (unchanged)
- *g → γ (unchanged)

Present 1sg: Add -ω ending (thematic present)

Predicted form: λέγω-ω

Attested form: ΛΕΓΩ ✓

Verification in other branches:

Latin (applying Italic sound laws):

- PIE *leǵ- → Proto-Italic *leg- (no change)
- Maintains as leg-
- Present 1sg: leg-ō

Attested: legō “I gather, read” ✓

Gothic (applying Grimm’s Law, Ringe 2006):

- PIE *leǵ- → Proto-Germanic *lag- (ǵ→g)
- Class IV strong verb: *ligan

Attested: -ligan “to lay” ✓

All predicted reflexes match attested forms. The reconstruction *leǵ- “gather; speak” accounts for Greek ΛΕΓΩ via regular sound changes and semantic development.

5. Discussion

The protocol demonstrated above operates through mechanical application of established IE principles. No interpretive leaps appear. Each step applies formalized rules.

Step 1 uses branch-specific morphological paradigms published in standard grammars. Step 2 queries existing etymological databases. Step 3 applies Brugmann's ablaut analysis. Step 4 considers semantic and morphological constraints documented in the inscriptional record. Step 5 reviews published reconstructions. Step 6 applies codified sound laws.

The process is reproducible. Given the same input and the same reference materials, independent analysts should arrive at the same candidate set.

The protocol does not require semantic operator values or novel theoretical constructs. It employs only the comparative method as practiced since the 19th century. What differs is systematization: we formalize the implicit process experts perform intuitively.

Several characteristics of the method deserve note:

First, the filtering is progressive. Each step narrows the candidate space using different evidence types. By combining phonological, morphological, and semantic constraints, the protocol converges even when individual evidence types underdetermine.

Second, the method is conservative. It retains candidates unless they violate established rules. Ambiguous cases remain ambiguous. The protocol identifies when evidence underdetermines and presents multiple candidates rather than forcing single answers.

Third, verification operates independently of reconstruction. Step 6 generates predictions testable against attested forms. This provides internal validation: correct reconstructions predict observed reflexes, incorrect ones do not.

Fourth, the finite search space makes exhaustive checking tractable. Because any skeleton maps to a limited set of PIE roots (typically under 100, often under 50), computational implementation could check all possibilities systematically.

The method has limitations. It assumes standard PIE reconstructions are broadly correct. It relies on published etymological databases. Substrate words, loans, and irregular formations may not follow the expected patterns. The protocol identifies when these cases arise—they generate no candidates or implausible ones—but cannot reconstruct them without additional evidence.

6. Conclusion

The Barnaby Protocol provides a systematic framework for etymological reconstruction operating entirely within established IE linguistics. The six-step process—affix stripping, skeleton mapping, vowel filtering, morphological filtering, candidate review, and reflex verification—applies codified sound laws, morphological paradigms, and comparative method principles to move from modern forms to PIE roots.

Demonstrated on ΛΕΓΩ, the protocol narrows from 67 initial candidates to a single reconstructed root through mechanical filtering. The predicted reflexes match attested forms across multiple branches.

The method is reproducible, testable, and falsifiable. It makes no theoretical innovations but systematizes existing practice. Historical linguists already perform these steps implicitly when researching etymologies. The contribution here is formalization: converting expert intuition into explicit procedure.

Applications include:

- Verification of published etymologies
- Investigation of disputed cases

- Training of new scholars
- Computational implementation
- Cross-linguistic database construction

The protocol demonstrates that etymology, despite its reputation for interpretive artistry, operates through rule-governed procedures amenable to systematization.

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Author Note: This protocol represents formalization of comparative method principles, not theoretical innovation. All linguistic laws and frameworks employed are standard in Indo-European studies.

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